Evaluating Agent Swarms: Token Budgets, Active Workers, and Baseline Choice

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Abstract

Agent swarm evaluation must distinguish increasing the total token budget, dividing a fixed ceiling, and changing coordination. We study these comparisons in 8,400 controlled trials on fresh arithmetic and graph-colouring tasks. At 4,096 completion tokens per worker, increasing independent width from one to four improves accuracy by 16.0 percentage points on both families while quadrupling the total ceiling. Splitting a fixed 4,096-token completion ceiling instead lowers graph accuracy from 56.0% to 18.5%; the arithmetic change is inconclusive. Primary comparisons do not establish an accuracy advantage over a sequential control matched on scheduled calls and nominal token ceiling. Ring communication provides no consistent gain. Eleven complementary scaling studies add 20,376 held-out trials across seven task families, including a fresh-task sixteen-worker confirmation that does not establish superiority over fixed-four execution. Candidate and request audits show that valid-answer recovery after initial failure depends on token allowance; they also reveal incomplete worker activation and 97 controlled-study attempts with unknown usage. Resource controls, output selection, and execution records thus qualify claims about agent scaling. The findings support evaluating specific allocation policies; equal realized compute advantages and general scaling laws remain unestablished from this work.

Keywords: Agent evaluation; Multi-agent systems; Test-time compute; Token budgets; Resource allocation; Coordination; Reliability; Reproducibility

> "All happy families are alike; each unhappy family is unhappy in its own way." —Leo Tolstoy, *Anna Karenina* (Tolstoy, 2014).
>
> *Adapted to this study:* All successful swarms share a valid outcome; unsuccessful swarms fail in different ways.

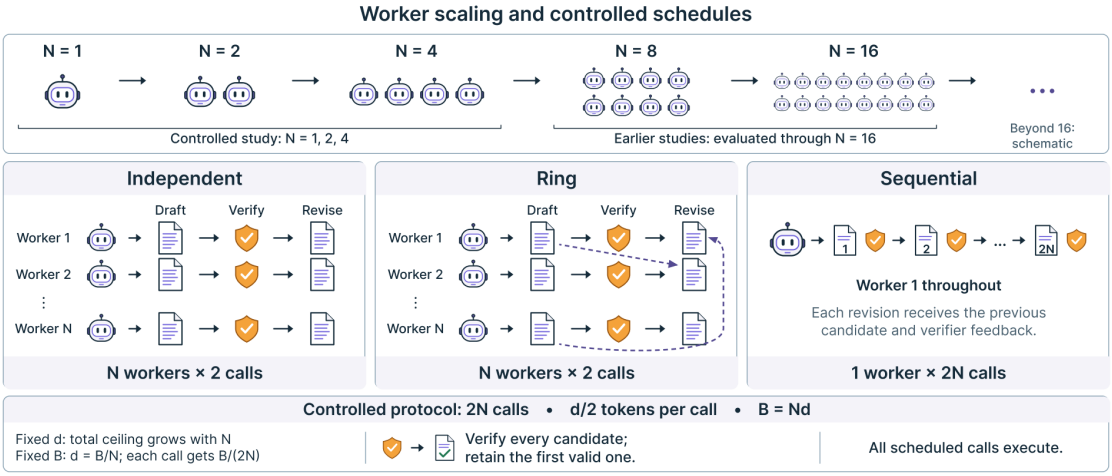


Figure 1 | Architectures and allocation: width N, per-worker completion allowance d, total ceiling $B = Nd$. All $2N$ calls execute with $d/2$ tokens per call. Controlled widths: 1, 2, 4; earlier studies: through 16. Larger widths are schematic.

1 Study and Findings

This ongoing study distinguishes sampling, aggregation, and debate (Brown et al., 2024; Wang et al., 2023; Du et al., 2023), with allocation controls for task and architecture dependence (Snell et al., 2024; Kim et al., 2026). Controlled widths are $N = 1, 2, 4$ (Figure 1); programming/ZebraLogic grids use $N = 1, 2, 4, 8, 16$, diagnostics $N = 1, 4$. Independent revision uses own drafts; ring revision receives predecessor information. The controlled sequential worker uses prior-candidate and verifier feedback. Task sources: BigCodeBench, MBPP/EvalPlus, ZebraLogic, BBEH, PlanBench (Zhuo et al., 2025; Austin et al., 2021; Liu et al., 2023; Lin et al., 2025; Kazemi et al., 2025; Valmeekam et al., 2023). Custom graph/Countdown tasks use deterministic checks; graph generation is inspired by Reasoning Gym (Stojanovski et al., 2025). Complementary studies retain 26,648 trials (6,272 calibration; 20,376 evaluation). Paired bootstrap resamples whole tasks with configurations and repetitions together, stratified where applicable (Efron, 1979; Field and Welsh, 2007); 95% CIs lack cross-study multiplicity correction.

All 148 wider cells (Figure 2) use more recorded tokens: 75 score lower, 24 equal, 49 higher. BigCodeBench Muse activation at $N = 16$ is 7.3%, 45.3%, and 93.5% at ceilings 32,768, 65,536, and 131,072. ZebraLogic $4/8/16$ minus fixed four is $+0.92$ pp (95% CI $[-0.58, 2.50]$), establishing neither superiority nor equivalence. Controlled fixed-total arithmetic: -24.0 pp, inconclusive.

At $B = 32{,}768$, independent BigCodeBench Muse passes 40/64 ($N = 1$), 36/64 ($N = 8$), and 14/64 ($N = 16$). Programming supplies 96 wider cells, 81 flat or worse; counts are not benchmark-balanced.

PlanBench's independent four-worker effect changes from -95.0 pp ($[-98.0, -91.5]$) to $+10.5$ pp ($[6.0, 15.5]$) as its ceiling doubles (higher-ceiling debit ratio: 4.06). BBEH ring effects change from -6.5 to $+4.5$ pp. Countdown's independent gains of 9.5–17.0 pp use 2.77–3.85 times the recorded tokens; its pooled ring effect is -5.0 pp ($[-8.5, -1.75]$). Our ring protocol is narrower than shared-workspace communication (Park et al., 2026).

Luna graph colouring's pooled ring effect is -2.0 pp ($[-4.0, -0.25]$). Seven Muse graph zeros are operational failures, not necessarily reasoning errors.

Resource audits complement failure taxonomies (Cemri et al., 2025). Unknown usage in 97 controlled attempts limits accounting; equal ceilings do not establish equal compute. Ongoing work tests matched resources and policy transfer; general scaling laws remain unestablished. Evidence and protocols: https://github.com/PluvioXO/agent-swarms-reproducibility.

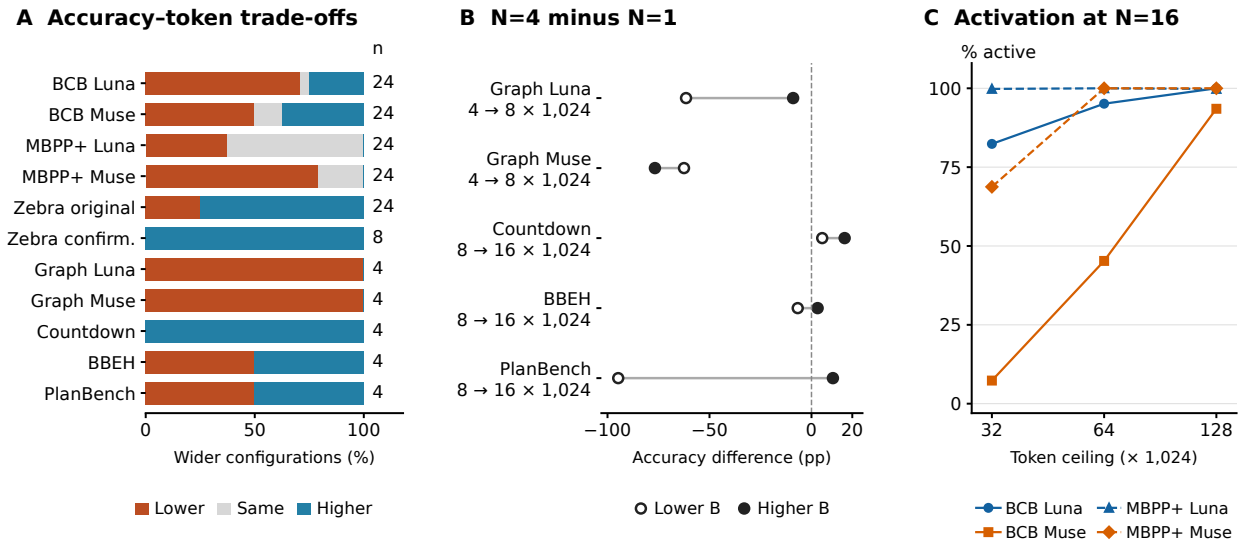


Figure 2 | (A) Wider-cell accuracy versus matched-ceiling $N = 1$ (148 configurations sharing tasks). (B) Four-worker effects at two ceilings, pooled over revision conditions. (C) Activation at sixteen requested workers.

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